

$$d) \neg \forall x P(x) : \neg (P(1) \wedge P(2) \wedge P(3) \wedge P(4) \wedge P(5))$$

$$e) \forall x (x \neq 3) \rightarrow P(x) \vee \exists x \neg P(x) : (P(1) \wedge P(2) \wedge P(4) \wedge P(5)) \vee (\neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4) \vee \neg P(5))$$

20) Suppose that the domain of the propositional function $P(x)$ consists of $-5, -3, -1, 1, 3$ and 5 . Express these statements without using quantifiers, instead using only negations, disjunctions and conjunctions.

$$a) \exists x P(x) : P(-5) \vee P(-3) \vee P(-1) \vee P(1) \vee P(3) \vee P(5)$$

$$b) \forall x P(x) : P(-5) \wedge P(-3) \wedge P(-1) \wedge P(1) \wedge P(3) \wedge P(5)$$

$$c) \forall x (x \neq 1) \rightarrow P(x) : P(-5) \wedge P(-3) \wedge P(-1) \wedge P(3) \wedge P(5)$$

$$d) \exists x ((x > 0) \wedge P(x)) : P(1) \vee P(3) \vee P(5)$$

$$e) \exists x (\neg P(x) \wedge \forall x (x < 0) \rightarrow P(x)) : (\neg P(-5) \vee \neg P(-3) \vee \neg P(-1) \vee \neg P(1) \vee \neg P(3) \vee \neg P(5)) \wedge (P(-5) \wedge P(-3) \wedge P(-1))$$

23) Translate in two ways each of these statements into logical expressions using predicates, quantifiers and logical connectives. First, let the domain consist of the students in your class and second, let it consist of all people.

a) Someone in your class can speak Hindi : Domain: Students in the class

$H(x) : x$ can speak Hindi $\exists x H(x)$

Domain: All people $S(x) : x$ is a student of your class

$\exists x (S(x) \wedge H(x))$

b) Everyone in your class is friendly. Domain: Students in the class

$F(x) : x$ is Friendly $\forall x F(x)$

Domain: All people $S(x) : x$ is a student of your class

$\forall x (S(x) \rightarrow F(x))$

c) There is a person in your class who was not born in California.

Domain: Students in the class. $C(x) : x$ was born in California.

$\exists x \neg C(x)$